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Title:TriShield Multilayer Authenticated Locker

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Theme: Security in IOT

Introduction

Tri-Shied Locker is an advanced security solution that combines PIN verification, OTP authentication, face recognition, and fingerprint verification to ensure secure access. It integrates an ESP32, Arduino Uno, Python, and Firebase for real-time communication, cloud-based user management, and access logging. A dedicated duress fingerprint allows an authorized user to appear to unlock the locker under threat while keeping the main vault secure and triggering a silent emergency alert. This multi-layered approach enhances security, reliability, and user safety compared to conventional locking systems.

Problem Definition

Current locker systems often rely on a single authentication method, making them vulnerable to unauthorized access and stolen credentials. They also lack secure credential storage, real-time access monitoring, and protection for users forced to unlock the locker under threat. This prototype addresses these issues by implementing multi-layer authentication, secure PIN storage using SHA-256 hashing, cloud-based access logging through Firebase, and a duress fingerprint feature that silently triggers an emergency alert while keeping the main vault secure.

Objectives

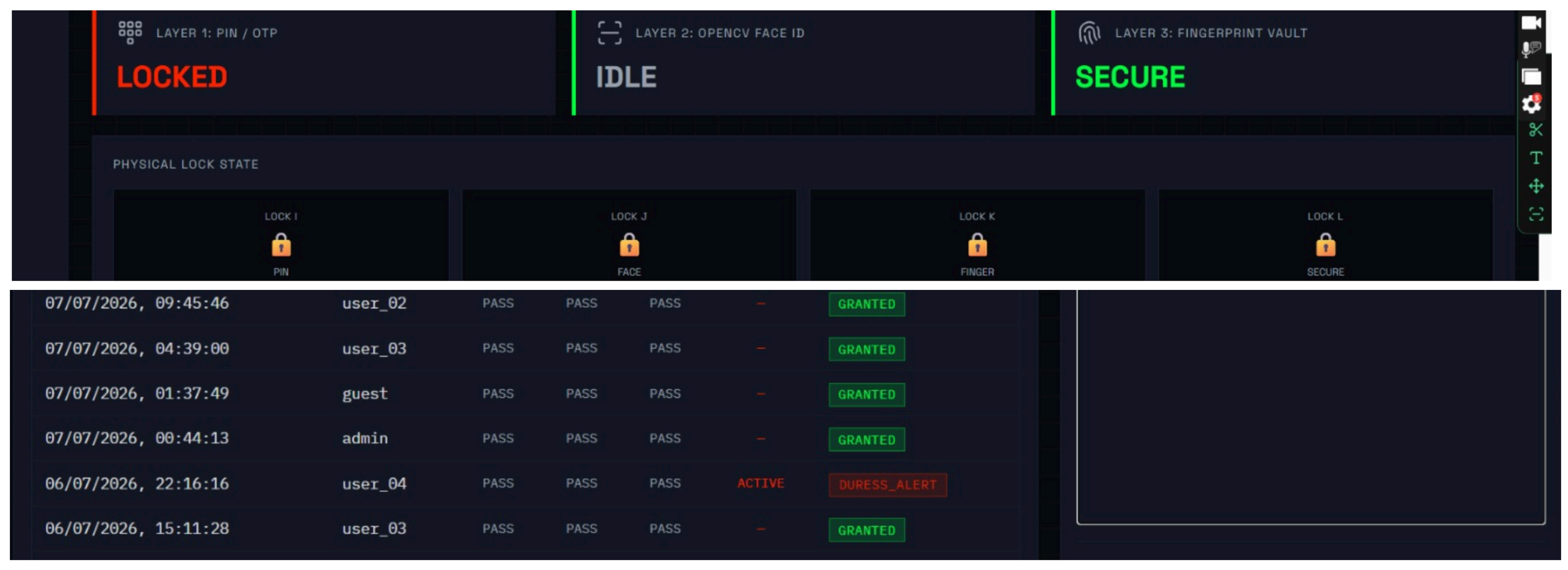
- To enhance locker security by implementing multi-layer authentication using a PIN, OTP, face recognition, and fingerprint verification.
- To securely store user credentials using SHA-256 hashing and manage user data through Firebase.
- To provide real-time cloud-based access logging and user management for monitoring authentication events.
- To improve user safety by incorporating a duress fingerprint feature that silently triggers an emergency alert while keeping the main vault secure.

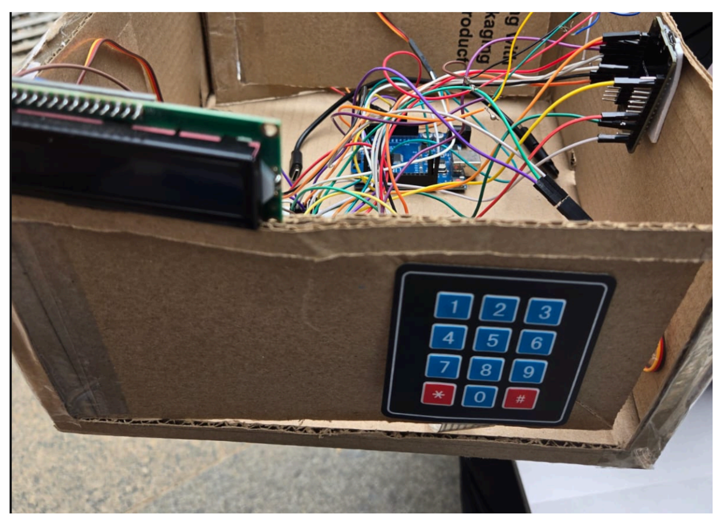
Methodology

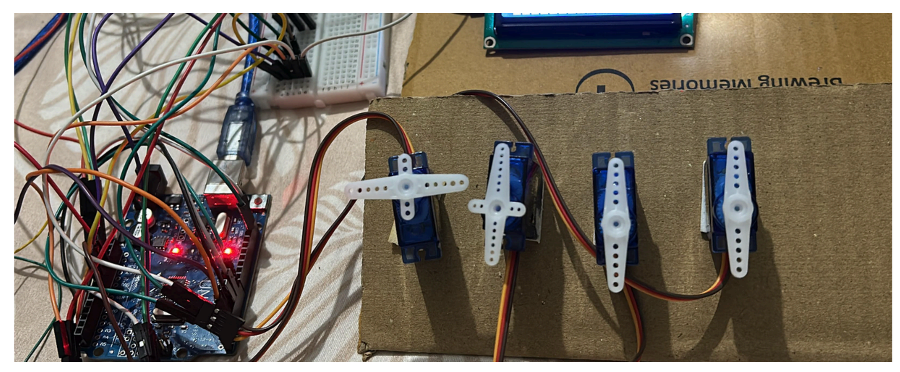
Tools used

- Arduino IDE – Programming the ESP32 and Arduino Uno.
- OpenCV – Capturing and processing facial images.
- Python – Developing the face recognition and Firebase integration modules.
- Firebase Realtime Database – Cloud storage for user credentials,.
- VS Code – Writing and managing Python and embedded system code.
- Firebase Admin SDK – Connecting the Python application with Firebase.
- Twilio API – Sending OTP via SMS.

Results and Discussions







Conclusions

- Successfully implemented a multi-layer authentication system using PIN, OTP, face recognition, and fingerprint verification.
- Enhanced security by storing user PINs securely using SHA-256 hashing and managing data through Firebase.
- Enabled real-time cloud-based user management and access logging for monitoring authentication events.
- Improved user safety with a duress fingerprint feature that silently triggers an emergency alert while keeping the main vault secure.
- Demonstrated a reliable and scalable smart locker solution by integrating embedded systems, biometrics, IoT, and cloud technologies.

Outcome of the work

Product Development: Successfully developed a functional prototype of a Smart Multi-Layer Biometric Locker System integrating PIN, OTP, face recognition, fingerprint authentication, Firebase-based cloud management, SHA-256 hashed credential storage, and a duress fingerprint safety mechanism.

References

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